Opacity as Commitment: Fragile Equilibria under Delegated Trading*

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Abstract

We study a model à la Kyle (1985) in which financial traders care not only about profits but also their reputation, i.e., a recruiter's perception of their skills. The model presents two self-fulfilling equilibria. One resembles Kyle's original equilibrium. In the other, a "reputation-driven" equilibrium, traders trade more aggressively to appear skilled, leading to poorer performance and higher price volatility. We show that (1) labor mobility in finance generates traders' reputation concerns, triggering market fragility and increasing wage inequality, (2) traders with different skill levels adopt different trading styles, and (3) skills in active management may not translate into performance.

JEL classification: G11, G12, G14, G23

Key words: informed trading, market reputation, trader skill, financial fragility, poaching, asset prices

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1 Introduction

Financial firms compete for skilled employees. As the boundaries between investment banking, commercial banking, and asset management blur, this competition intensifies. Following the 2007-2009 crisis, pressures to reduce compensation and the resulting challenges in retaining talent have brought poaching practices into focus.[1] Poaching, or more generally labor mobility, raises concerns among financial specialists about their reputation within the industry. How do these concerns influence their trading strategies and skill acquisition? How do they interact with asset prices and market liquidity?

To address these questions, we study a model inspired by Kyle (1985), where traders, privately informed about their skill, trade risky assets anonymously with market makers. We define skill as the trader-specific ability to uncover more information about asset payoffs than the market. The more skilled a trader is, the larger the potential mispricing she can identify, and thus, the greater the trading profits she can potentially earn. Each trader selects an asset based on her skill and strategically submits a market order to the market maker for that asset, who sets the price after observing the aggregate order flow. A recruiter attempts to infer each trader's skill from the observed price and payoff of the asset chosen by the trader. The trader values not only trading profits but also her reputation, i.e., the recruiter's estimate of her skill. In a simple one-period model presented in Section 2, each trader derives an exogenous benefit from her reputation, which we model as an increasing quadratic function of reputation. In Section **??**, we extend the model to a two-period setting, providing a microfoundation for the reputation benefit. In this extension, each trader randomly encounters a recruiter who compensates her for her skill based on his estimate. In equilibrium, the quadratic form

[1]According to *The Wall Street Journal*, poaching is a common practice in the sophisticated capital markets industry, occurring in both bull and bear markets ("Greed is Good," February 2009), and employees at all levels are being poached ("Employee Poaching Returns to Wall Street," May 2010). *The New York Times* ("A Mad Scramble for Young Bankers," July 2014) and *Bloomberg* ("Saturdays Off Lose to Cash as Buyout Firms Poach Junior Bankers," March 2014) highlight the buy-side's (private equity, hedge funds) recruitment of young talent from investment banking amid a changing financial industry.

assumed in Section 2 arises endogenously.

The model yields two stable, self-fulfilling equilibria, one of which is similar to that of Kyle (1985), where reputation is irrelevant. In the other, the "reputation-driven" equilibrium, traders trade more aggressively to appear skilled. This strategy increases trading volume, price volatility, and market liquidity but undermines trading profits by revealing too much private information to the market maker. Intuitively, each trader can boost her reputation by placing a large order, either buy or sell, because such an action makes her appear like a skilled trader who has identified an asset with significant mispricing. In equilibrium, the recruiter accurately anticipates the trader's strategy, so the recruiter's learning is not manipulated; nevertheless, the trader may trade aggressively depending on the market's belief. To illustrate the mechanism, suppose the recruiter believes that the trader trades more aggressively than the profit-maximizing level. If she trades less aggressively to increase profits, the recruiter (incorrectly) infers that she is less skilled. To avoid the reduction in her reputation benefit, she conforms to the recruiter's belief and indeed trades very aggressively.

The self-fulfilling property of our equilibria implies that a mere shift in investor beliefs, unaccompanied by changes in fundamentals, can trigger a transition from the standard Kyle-like equilibrium to the reputation-driven one, where price volatility is high and trading performance is relatively poor. We interpret this as *fragility* in financial markets.[2] Key to our multiple equilibria and fragility results are the trader's conflicting motives for revealing skill: on one hand, she wishes to hide her skill from the market maker to earn trading profits, but on the other hand, she wants to show it off to the recruiter to obtain the reputation benefit. If the trader's concern for reputation is very weak (very strong), the first (second) motive dominates, and the Kyle-like (reputation-driven) equilibrium holds uniquely. However, if the concern is moderately strong, either equilibrium may arise depending on the market's self-fulfilling beliefs. Therefore, the model predicts that fragility may occur

[2] As in Greenwood and Thesmar (2011), we define a market as fragile if it is susceptible to non-fundamental demand shifts, such as those caused by changes in investor beliefs.

when poaching is (moderately) active and there is high demand for financial skill, as these situations increase the likelihood of future recruitment and lead to heightened reputation concerns.

Our results shed light on how traders' underlying skills relate to their trading styles and contribute to financial fragility. The model predicts that high-skilled traders adopt low-intensity, profit-driven strategies, while those with lower skills engage in aggressive, risk-seeking behavior to gain reputation. Traders with intermediate skill levels, facing multiple equilibria, may switch between profit-seeking and reputation-driven strategies based on market beliefs, thereby contributing to market fragility. This behavior aligns with real-world observations during financial crises, such as momentum crashes and sudden reversals, which occur even without fundamental shocks.

The model has implications for compensation in the finance industry. In the reputation-driven equilibrium, traders' wages exhibit greater inequality among high- and low-skilled traders compared to those in the Kyle-like equilibrium, as they trade aggressively and reveal their skill more than they do in the Kyle-like equilibrium. This allows the recruiter to better assess each trader's skill and, therefore, to offer wages that more accurately reflect their abilities. This result is consistent with the findings by Philippon and Reshef (2012) and Célérier and Vallée (2019), who document increasing returns to talent and their significant contribution to income inequality in the finance industry.

The aggressive trade in the reputation-driven equilibrium also causes traders to reveal excessive information to the market maker, resulting in poorer trading performance than in the Kyle-like equilibrium. Consequently, high-skilled traders in the reputation-driven equilibrium may underperform low-skilled traders following the Kyle-like equilibrium. This result offers a novel explanation for the puzzling relationship (or the lack thereof) between trader skill and persistent performance, as documented and analyzed in the literature (e.g., Busse, Goyal, and Wahal, 2010; Berk and Green, 2004).

In Section **??**, we study traders' endogenous skill acquisition. We show that, due to the multiplicity of equilibria, each trader may intentionally acquire a "middling" level of skill, which is too high or too low compared to the level she would acquire if she knew which type of equilibrium would prevail. Ex-post, in the Kyle-like equilibrium, where profits are prioritized over reputation, the acquired skill level is insufficient. Conversely, in the reputation-driven equilibrium, where reputation outweighs profit motives, the skill level is excessive. Ex-ante, without knowing which equilibrium will emerge, the trader optimally chooses a middling skill level to prepare for both scenarios. This finding aligns with real-world observations that some finance professionals tend to pursue broader business qualifications rather than specializing in areas like advanced computer science or pursuing higher academic degrees such as a PhD.

Our paper is related to the literature on fragility in financial markets. Cespa and Vives (2015) is most closely related, as they also find multiple equilibria in an asymmetric information trading model. The key ingredients of their model are short-term investor horizons and persistent liquidity trading, which generate strategic complementarities. In Froot, Scharfstein, and Stein (1992), strategic complementarities among short-term traders are also important for the emergence of multiple equilibria. Unlike these studies, our model does not feature short horizons; instead, we obtain fragility from traders' forward-looking concerns about future recruitment opportunities. In Cespa and Foucault (2014), learning from other assets amplifies shocks and causes liquidity crashes. Although the mechanism is different, our model can also result in liquidity crashes when the economy switches between equilibria due to non-fundamental shocks.

Our paper also connects to the literature on trading frenzies, such as Veldkamp (2006) and Goldstein, Ozdenoren, and Yuan (2013), in which strategic complementarities play a key role. In Veldkamp (2006) frenzies originate from complementarities in information markets, while in Goldstein, Ozdenoren, and Yuan (2013) strategic complementaries, triggered by a feedback effect from financial markets to a firm's real investment, lead to coordinated aggressive trading. By contrast, in our reputation-driven equilibrium, each trader trades aggressively out of fear of being perceived as less skilled by the recruiter who believes that she would trade aggressively.

Lastly, our work contributes to the broader understanding of the role of asymmetric information about skills in financial markets. DiMaggio (2015), Malliaris and Yan (2021), and Bijlsma, Boone, and Zwart (2018) derive implications for the risk-taking behavior of agents when ability is private information but is reflected in performance. We also link trader skill to private information and asset markets; however, unlike these papers, our focus is on deriving pricing implications. Guerrieri and Kondor (2012) show that career concerns amplify price volatility, similar to our findings. In Dasgupta (2006), career concerns lead to churning (trading without private information), and in Dasgupta and Prat (2008), they lead to herding (ignoring private information and trading like everyone else). These three papers are closely related to ours in that they study career concerns and endogenous asset prices, but they do not analyze fragility, which is the main focus of our paper.

The rest of the paper proceeds as follows. Section 2 presents a baseline model with exogenous reputation concerns, and Section 3 studies the equilibrium. Section ?? endogenizes reputation concerns by introducing a two-period model and presents the implications of our results. Section ?? analyzes traders' skill acquisition. The Online Appendix contains all proofs for the theoretical results.

2 Model

This section presents a one-period trading model à la Kyle (1985) with delegated investments. The economy consists of a risk-neutral *fund manager*, competitive risk-neutral *market makers*, *noise traders*, and $n \geq 2$ risk-neutral *investors*. The manager leverages her skill (defined later) to discover a single risky asset with potential mispricing, raises funds from investors, and trades the asset with market makers who set the price p. The remaining funds are allocated to risk-free savings with a zero interest rate.[3]

[3]We abstract away from information acquisition by individual investors. Economies of scale would support this assumption as a natural equilibrium outcome, in which investors do

Skill. The manager can uncover more information about the asset payoff than market makers, and we interpret it as her skill as follows. At the beginning of the period, the manager draws a number s from a normal distribution, $N(0, \omega_s)$, where s is her private information. Leveraging s, she identifies an asset whose payoff is $\delta = \bar{\delta} + s$ with $\bar{\delta}$ representing a publicly known mean payoff. Since market makers do not observe s, the prior mean of δ from their perspective is $\mathrm{E}[\delta] = \bar{\delta} + \mathrm{E}[s] = \bar{\delta}$, which is biased by s from the manager's perspective. As market makers would set a price $p = \mathrm{E}[\delta] = \bar{\delta}$ in the absence of traders, s can be viewed as the asset's "potential mispricing."[4] Indeed, as shown later, the manager profits by buying the asset if $s > 0$ (underpriced) and selling it if $s < 0$ (overpriced); the larger the $|s|$, the larger the profit. Thus, we refer to $|s|$ as the manager's *realized* asset-selection skill.[5]

Moreover, the variance of the realized skill, $\mathrm{Var}[s] = \omega_s$, controls the shape of the underlying distribution from which the manager draws s. A large ω_s represents a distribution with a wider scope, making the manager more likely to acquire a high realized skill level, $|s|$. Therefore, we refer to ω_s as the manager's *skill*. In practical terms, ω_s can reflect factors such as the manager's investment in higher education in finance and economics, which broadens the range of skills available to her.

Investment. We model the fundraising and trading stages as a simultaneous game with rational expectations. On the one hand, the manager decides on the investment into the risky asset, anticipating the investors' and market makers' reactions, and places a market order for x units to market makers. In the financial market, noise traders also submit random order $u \sim N(0, \omega_u)$ exogenously. Market makers observe the aggregate order flow, $x + u$, and set

not acquire the necessary skills and information, but rather remain uninformed and invest through a manager (e.g., Gârleanu and Pedersen, 2018). They do not directly participate in the market due to the lack of informational advantages over market makers.

[4] s is not the *actual* mispricing because market makers will take into consideration the presence of the informed manager; thus, p will deviate from $\bar{\delta}$ in equilibrium.

[5] This formalization of skill aligns with findings in the empirical studies that attribute fund managers' performance to stock-picking skills (e.g., Wermers, 2000; Kosowski, Timmermann, Wermers, and White, 2006).

the semi-strong efficient price to earn zero expected profit, that is,[6]

$$p = \mathrm{E}[\delta | x + u]. \tag{2.1}$$

In raising funds from investors, the manager disseminates a noisy signal about her position to investors, $x + e$, where e represents noise that follows a normal distribution $e \sim N(0, \omega_e)$. This signal is delivered through private communication between the manager and her customer investors and conveys imperfect information about the fund's prospective risky holdings. It is informative about the manager's skill and the return of the fund, and investors use it in deciding on fund allocations. We interpret ω_e as the *opacity* of the fund. We first consider the case with a perfectly transparent fund ($\omega_s = 0$) as our baseline model, while Section XXX deals with general opacity ($\omega_s > 0$). Section XXX endogenizes ω_s through the manager's choice, e.g., she may obfuscates information by making trading strategies complex and incomprehensible to uninformed investors. Upon observing the fund information, $x + e$, investor i allocates m_i dollars to the fund. In aggregate, the fund's asset under management (AUM) becomes $M = \sum_{i=1}^{n} m_i$. The fund's fee structure is proportional to the total profits: the manager takes ϕ fraction of the total fund profit, while the remaining $1 - \phi$ fraction is distributed to individual investors according to their contributions to the fund, m_i/M.[7]

Maximization problem. With the trading profit from the risky investment being $\pi \equiv (\delta - p)x$, the total fund proceeding is expressed as

$$y = \pi + M. \tag{2.2}$$

[6] Assume that, on the equilibrium path, one competitive market maker trades the asset. As in Kyle (1985), competitively many market makers exist off the equilibrium path, ensuring a break-even price in the equilibrium.

[7] Introducing a fixed fee does not change our qualitative results, as long as the performance-based fee also exists, $0 < \phi < 1$.

The fund manager maximizes her fee income by controlling her investment strategy:

$$\max_x \mathrm{E}\left[\phi y |\, s\right]. \tag{2.3}$$

Conversely, investor i maximizes her return from investment by controlling her fund contribution:

$$\max_{m_i} \mathrm{E}\left[\frac{m_i}{M}(1-\phi)y - m_i | x + e\right]. \tag{2.4}$$

Definition of equilibrium. Let $p(x+u)$ represent market makers' pricing strategy, $x(s)$ denote the manager's trading strategy, and $\{m_i(x+e)\}_{i=1}^n$ denote investors' fund allocations. An equilibrium consists of $p(\cdot)$, $x(\cdot)$, and $\{m_i(\cdot)\}_{i=1}^n$ such that $p(\cdot)$ satisfies (2.1) given $x(\cdot)$, $x(\cdot)$ solves (2.3) given $p(\cdot)$ and $\{m_i(\cdot)\}_{i=1}^n$, and each $m_i(\cdot)$ solves (2.4) given $p(\cdot)$ and $x(\cdot)$.

3 Equilibrium

In this section, we search for a linear equilibrium under transparency ($\omega_s = 0$), in which each trader's market order x is linear in her skill, s. Proofs for the analytical results are provided in the Online Appendix.

3.1 Trading Strategy

Conjecture and later verify that the manager's equilibrium trading strategy is

$$x(s) = \beta s, \tag{3.1}$$

where $\beta > 0$ is an endogenous constant determined later. Conjecture (3.1) states that the manager buys (sells) the asset if her s is positive (negative), where the order size is proportional to her skill $|s|$; the more skilled the manager, the more aggressively she trades.[8] The scale factor β is referred to as the *trading intensity*.

[8] This conjecture follows from the original Kyle model where an informed trader trades based on the difference between her belief and that of the market maker, i.e., $\mathrm{E}[\delta|s] - \mathrm{E}[\delta] = s$.

3.2 Execution Price

Market makers decide on the price of each asset by extracting information about the asset's payoff from the aggregate order flow, $x+u$. Anticipating the trading strategy (3.1), the price in equation (2.1) can be rewritten as

$$p = \bar{\delta} + \lambda(x+u), \tag{3.2}$$

where

$$\lambda \equiv \frac{\beta\omega_s}{\beta^2\omega_s + \omega_u} \tag{3.3}$$

is "Kyle's lambda," a measure of the trader's price impact.[9] Following the literature, we define a market to be *liquid* if λ is small.

3.3 Investment Delegation

Investor i's optimization problem in (2.4) is rewritten as

$$\max_{m_i} \frac{m_i}{M}(1-\phi)\mathrm{E}\left[\pi|x+e\right] - \phi m_i, \tag{3.4}$$

leading to the following optimal delegation:

$$m_i = \left(\frac{1-\phi}{\phi}\mathrm{E}[\pi|x+e]\sum_{j\neq i} m_j\right)^{\frac{1}{2}} - \sum_{j\neq i} m_j. \tag{3.5}$$

The cost of delegating investment is linear in the dollar value of the stake, as represented by the second term of (3.4), while the marginal benefit of an additional stake is diminishing, as is captured by the first term. Hence, given the total fund allocation by other investors, $\sum_{j\neq i} m_j$, investor i's optimal delegation is determined by the FOC as (3.5).

We focus on the symmetric equilibrium where all investors provide the same

[9] From (2.1), we have $p = \mathrm{E}[\delta|x+u] = \mathrm{E}[\mathrm{E}[\delta|x+u, s]|x+u] = \mathrm{E}[\bar{\delta}+s|x+u]$. Since market makers believe that the manager follows (3.1), s and $x+u$ are both normally distributed from market makers' perspective. Thus, $p = \bar{\delta} + \mathrm{E}[s] + \frac{\mathrm{Cov}[s,x+u]}{\mathrm{Var}[x+u]}(x+u-\mathrm{E}[x+u]) = \bar{\delta} + \frac{\mathrm{Cov}[s,\beta s+u]}{\mathrm{Var}[\beta s+u]}(x+u-\mathrm{E}[\beta s+u]) = \bar{\delta} + \frac{\beta\omega_s}{\beta^2\omega_s+\omega_u}(x+u)$.

amount of capital to the fund. By imposing $m_i = m_j$ for all $i, j \in \{1, 2, \cdots, n\}$, (3.5) is reduced to

$$m = \frac{n-1}{n^2}\frac{1-\phi}{\phi}\mathrm{E}[\pi|x+e]. \tag{3.6}$$

It also yields the total asset under management (AUM) of the fund as

$$M \equiv nm = \frac{n-1}{n}\frac{1-\phi}{\phi}\mathrm{E}[\pi|x+e]. \tag{3.7}$$

In what follows, we denote $\frac{n-1}{n}\frac{1-\phi}{\phi} = \theta$. Individual and the total fund provisions are proportional to the expected value of trading profits conditional on the fund information disseminated to investors, $\mathrm{E}[\pi|x+e]$. This is a natural consequence of the performance-based fee structure, as it makes the per-unit return of fund delegation proportional to the trading profit.

By incorporating the trading strategy in (3.1) and the pricing rule in (2.1), both the trading quantity, x, and the profit margin (i.e., informational advantages), $\mathrm{E}[\delta - p|s]$, are proportional to the realized skill level, $|s|$, making the profit a quadratic function of s. When the fund information is perfectly transparent ($\omega_e = 0$), investors back out the realized skill level directly from the signal, $s = \frac{x}{\beta}$, so that the conditional expectation leads to the following estimate:

$$\mathrm{E}\left[\pi|x+e\right] = (1-\lambda\beta)\beta\mathrm{E}[s^2|s = x/\beta] = \frac{1-\lambda\beta}{\beta}x^2. \tag{3.8}$$

Together with the total fund provision in (3.7), the above estimate implies that the delegated investment may generate an additional incentive for the manager to control her trading strategy, x. Namely, the manger can positively influence the total fund flow by inflating the trading quantity, as it persuades investors of the manager's high realized skill level.

3.4 Manager's Optimization

By incorporating the total fund flow ([3.7] and [3.8]), the manager's problem in (2.3) is re-written as

$$\max_x \phi \left[sx - \lambda x^2 + \theta \frac{1-\lambda\beta}{\beta} x^2 \right], \tag{3.9}$$

where the first two terms represents the fee income from the trading profit and are essentially identical to the original Kyle (1985) model. The third term arises due to the fund inflow and is determined by the quadratic size of risky investments. By incorporating the pricing rule (2.1) and taking the FOC, the optimal trading strategy involves

$$x = \frac{1}{2} \frac{\beta}{\lambda\beta - \theta(1-\lambda\beta)} s. \tag{3.10}$$

This strategy is consistent with our guess in (3.1) when $\beta = \frac{1}{2}\frac{\beta}{\lambda\beta-\theta(1-\lambda\beta)}$, leading to the following lemma:

Lemma 3.1. *When the fund information is transparent ($\omega_s = 0$), the equilibrium trading strategy and the pricing rules are specified by*

$$\beta_0 = \sqrt{(1+2\theta)\frac{\omega_u}{\omega_s}}, \;\; \lambda_0 = \frac{1}{2(1+\theta)}\sqrt{(1+2\theta)\frac{\omega_s}{\omega_u}}. \tag{3.11}$$

As is evident from problem (3.9), parameter θ captures the manager's additional incentive to control her risky investment arising from the fund delegation, and the equilibrium reduces to the original Kyle (1985) as $\theta \to 0$. This case highlights the manager's tradeoff in deciding on her trading intensity: while she seeks to exploit her informational advantage by trading more intensively, it reveals information to market makers and raises the price impact, reducing the trading profit. Without fund delegation ($\theta = 0$), the equilibrium β is determined to balance these competing effects. Conversely, when θ increases, the share of the total fund flow becomes larger in the manager's utility, and she becomes more concerned about it. As shown in (3.7) and (3.8), inflating the trading quantity benefits the manager by persuading investors of high $|s|$ and inducing a larger fund flow. Therefore, β becomes an increasing function of θ. XXXThe price impact, λ, decreases with θ, as the aggressive trading reveals too much private information to market makers.

[XXX Implication of Delegation XXX]

$$F(\beta) = \tag{3.12}$$

With this $\hat{s}$, the trader's objective in equation (**??**) can be written as a simple quadratic function of x, as presented in the following lemma.

Lemma 3.2 (Trader's problem)**.** *Each trader's maximization problem is*

$$\max_x \ sx - \lambda x^2 + \theta\xi^2 \left(\gamma^2 x^2 + 2\gamma(1-\gamma)sx\right). \tag{3.13}$$

The first term of equation (3.13) is the trader's expected trading revenue. The second term represents the trading cost resulting from her price impact, which is increasing in the size of her trade, $|x|$. The third term corresponds to her reputation benefit, which also increases with $|x|$ because she can potentially inflate her reputation $|\hat{s}|$ by placing a large order (either buy or sell) and influencing the price that the recruiter observes.

The unique solution to (3.13) (given β) is

$$x = F(\beta)s, \tag{3.14}$$

where

$$F(\beta) \equiv \frac{1 + 2\theta\xi^2\gamma(1-\gamma)}{2\left(\lambda - \theta\xi^2\gamma^2\right)}. \tag{3.15}$$

$F(\cdot)$ is a nonlinear function of β because λ, ξ, and γ are functions of β.

3.5 Equilibrium

Each trader optimally chooses (3.14) given that the market maker and the recruiter believe that she follows strategy (3.1). In equilibrium, their beliefs must be correct, that is, (3.1) and (3.14) must be consistent with each other, requiring the following condition:

$$\beta = F(\beta). \tag{3.16}$$

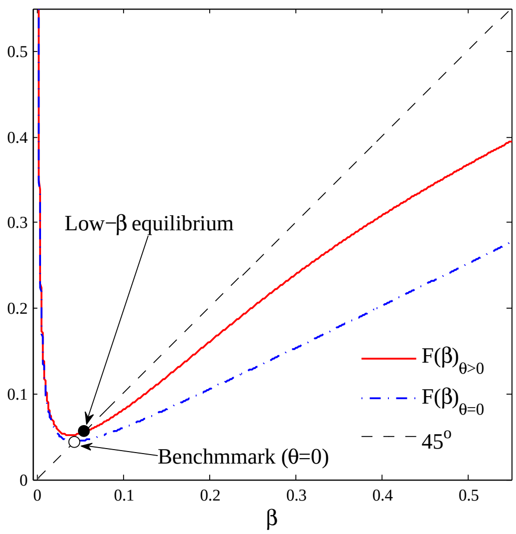
That is, the equilibrium levels of β are determined at the fixed points of $F(\cdot)$. Having identified β, the equilibrium levels of x and p are determined by equations (3.1) and (3.2), respectively. In what follows, we first illustrate the determination of equilibrium β through a numerical example, followed by an analytical proposition that formally establishes the equilibria.

Figure 1 illustrates the determination of β for three different levels of θ. The intersections of $F(\beta)$ (the solid line) and the 45-degree line yield the equilibrium levels of β. As a benchmark, we also plot $F(\beta)$ with $\theta = 0$ (the dash-dotted line), which corresponds to the model of Kyle (1985) where reputation concerns are irrelevant. The trading intensity in the benchmark equilibrium is denoted as $\beta_{\theta=0}$.

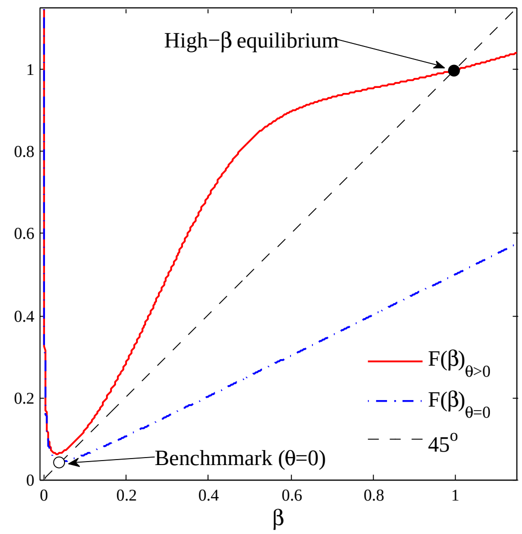
Panel (a) of Figure 1 shows that if θ is small, there is a unique "low-β equilibrium" where β is close to $\beta_{\theta=0}$. Panel (b) shows that a large θ leads to a unique "high-β equilibrium," in which β is much larger than $\beta_{\theta=0}$.[10] Panel (c) shows that multiple equilibria arise for an intermediate level of θ. Specifically, the high-β and the low-β equilibria are stable, as $F(\beta)$ crosses the 45-degree line from above, whereas the one with an intermediate β is unstable, as $F(\beta)$ crosses the line from below.[11] In what follows, we focus only on the stable equilibria.

[10]Formally, we define an equilibrium to be low-β (high-β) if $\beta < \tilde{\beta}$ ($\beta > \tilde{\beta}$), where $\tilde{\beta}$ is defined as square root of the unique solution to equation (**??**) in Lemma **??** of Online Appendix **??**. In the context of Figure 1, a low-β equilibrium is given by a fixed point of function F at which F is convex and U-shaped, while a high-β equilibrium is given by a fixed point that is larger than the first inflection point of F. Using this definition, Kyle's original equilibrium ($\theta = 0$) is classified as a low-β equilibrium.

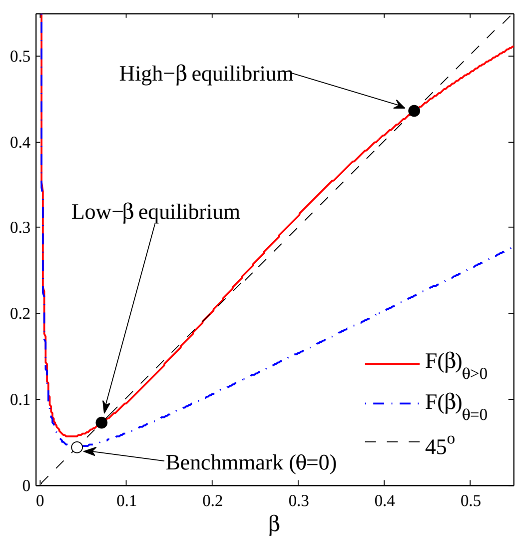
[11]We define an equilibrium to be stable (unstable) if the corresponding β is a stable (unstable) fixed point of $F(\cdot)$, that is, $|F'(\beta)| < 1$ (> 1). A stable equilibrium is robust to a small perturbation to the agents' belief about β, meaning that the economy converges back to the original equilibrium through the best-response dynamics of the agents' beliefs.



(a) Small θ ($\theta = 0.2$) $\Rightarrow$ unique low-β equilibrium.



(b) Large θ ($\theta = 0.55$) $\Rightarrow$ unique high-β equilibrium.



(c) Intermediate θ ($\theta = 0.33$) $\Rightarrow$ multiple high-β & low-β equilibria.

Figure 1: Determination of β

Note: The figure plots $F(\beta)$ for different values of θ. The equilibrium values of β are determined at the fixed points of $F(\cdot)$. The parameter values used in the figures are $\omega_s = 50$, $\omega_\delta = 1$, and $\omega_u = 0.1$.

Impact of reputation concerns. To see the intuition behind the different equilibrium patterns presented in Figure 1, note that the trader faces two conflicting motives when trading. On the one hand, she seeks to avoid trading too aggressively on her private information, as doing so would move the price excessively and reduce her trading profits. As in Kyle (1985), this motive leads the trader to decrease β. On the other hand, she wishes to trade intensively to signal her high skill to the recruiter, aiming to establish a strong reputation. This motive encourages her to increase β. In summary, she faces a tradeoff between "hiding" her skill from the market maker and "showing it off" to the recruiter. The degree of reputation concern, θ, governs this tradeoff.

For small θ (panel (a)), the hiding motive dominates the showing-off motive, leading to a unique low-β equilibrium. In this equilibrium, the recruiter believes that the trader's trading intensity is relatively low. Knowing this, the trader could inflate her reputation by placing an order larger than what the recruiter anticipates. However, she would refrain from doing so because it would cause a large price impact, reducing her trading profits. Consequently, the equilibrium β is close to $\beta_{\theta=0}$, meaning that it is a "Kyle-like equilibrium."

For large θ (panel (b)), the showing-off motive prevails, resulting in a unique high-β equilibrium. In this equilibrium, the trader is willing to move the price substantially, undermining her profits to appear skilled. Importantly, the recruiter's learning is not manipulated, as his belief about the trader's trading strategy is correct. Nonetheless, the trader places a large order because doing otherwise would induce the recruiter to incorrectly estimate that she is less skilled, thereby deteriorating her reputation. She trades aggressively solely to conform to the recruiter's belief. To emphasize this mechanism, we call this equilibrium the "reputation-driven equilibrium."

For intermediate levels of θ (panel (c)), there is no clear dominance between the two motives, hence both the low-β and the high-β equilibria are supported. Each equilibrium is associated with a self-fulfilling belief of the market maker and the recruiter.

The following proposition and Figure 2 summarize the above discussion and provide an analytical characterization of the equilibria (the proof is provided

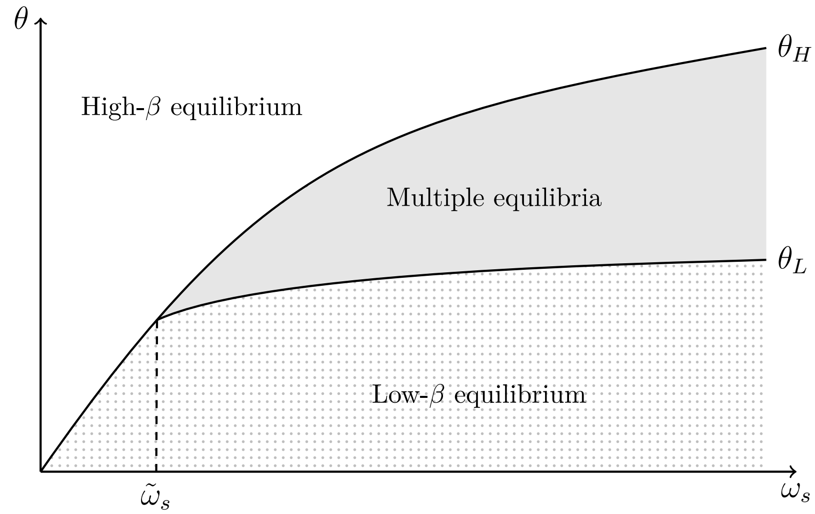


Figure 2: Equilibrium Characterization with Exogenous θ

Note: This figure illustrates the equilibrium states using the reputation concern, θ, and the skill potential, ω_s.

in Online Appendix **??** as a special case of Proposition **??**).

Proposition 3.1 (Equilibrium characterization)**.** *There exists a critical value of the skill potential, $\tilde{\omega}_s > 0$, and two thresholds for reputation concern, θ_L and θ_H $(\geq \theta_L)$, such that, $\theta \geq \theta_H$ leads to a unique high-β equilibrium, while $\theta \leq \theta_L$ leads to a unique low-β equilibrium. Furthermore,*

(i) if $\omega_s \leq \tilde{\omega}_s$ then $\theta_H = \theta_L$, hence the equilibrium is unique; and

(ii) if $\omega_s > \tilde{\omega}_s$ then $\theta_H > \theta_L$, and $\theta \in (\theta_L, \theta_H)$ leads to multiple equilibria.

Impact of skill potential. Proposition 3.1 and Figure 2 indicate that, for a given θ, a low skill potential ω_s supports only the high-β equilibrium, while increases in ω_s can generate the low-β equilibrium, potentially leading to multiple equilibria.[12] To understand intuition, the tradeoff between hiding and showing-off motives, as explained above, is key. When ω_s is small, the market maker anticipates that the trader has low skill and, due to limited adverse

[12]Lemma **??** in Online Appendix **??** analytically characterizes the upward-sloping θ_H and θ_L in relation to ω_s, as presented in Figure 2.

selection, sets a low price impact λ. In this case, the trader's hiding motive is less important. Consequently, the showing-off motive prevails, prompting her to trade aggressively and leading to the high-β equilibrium. Conversely, when ω_s is large, the market maker sets a high λ, and the hiding motive becomes more influential, making the low-β equilibrium more likely to be realized.

Overall, the type of equilibrium depends on the combination of θ and ω_s, as illustrated in Figure 2. Small θ and large ω_s, both of which amplify the trader's hiding motive, lead to the low-β equilibrium. Conversely, large θ and small ω_s, which both intensify her showing-off motive, result in the high-β equilibrium. However, with a combination of large θ and large ω_s, the result is ambiguous, as the two motives offset each other. In this case, either the high-β or the low-β equilibrium may be realized, depending on the market's self-fulfilling beliefs.

Equilibrium comparison. Comparing the two stable equilibria, does the trader perform better in one of them? What are the implications for market liquidity, asset prices, and learning about asset-selection skills?

Corollary 3.1 (High-β vs. low-β equilibrium)**.** *In the high-β equilibrium, relative to the low-β equilibrium,*

(i) the market liquidity is higher, that is, λ is lower;

(ii) the trader's expected trading profit given the realized skill, $\mathrm{E}[(\delta-p)x|s] = \frac{\beta\omega_u s^2}{\beta^2\omega_s+\omega_u}$*, is lower;*

(iii) the price informativeness, $\tau \equiv \frac{1}{\mathrm{Var}[\delta|p]} = \frac{\beta^2\omega_s+\omega_u}{\omega_\delta(\beta^2\omega_s+\omega_u)+\omega_u\omega_s}$*, is higher;*

(iv) the price volatility, $\mathrm{Var}[p] = \frac{\beta^2\omega_s^2}{\beta^2\omega_s+\omega_u}$*, is higher; and*

(v) the recruiter's estimate of s is more precise, that is, the posterior variance of s, $\hat{\omega}_s = \frac{\omega_s\omega_u\omega_\delta}{\omega_s(\beta^2\omega_\delta+\omega_u)+\omega_u\omega_\delta}$*, is lower.*

Statement (i) of Corollary 3.1 holds because price impact λ is decreasing in β. Since the high-β equilibrium emerges only if θ is relatively large, the statement also implies that a higher reputation concern leads to higher market liquidity. The intuition is as follows. Relative to the low-β equilibrium, in the high-β equilibrium the trader trades more aggressively and thus reveals more private information on s to the market maker. From the market maker's perspective, the information is "exaggerated" in that the order size is too large relative to the true $|s|$. Thus, the market maker attempts to partially "undo" the trader's information revelation by incorporating less of it into the price. This mechanism results in a reduced price impact.

The intuition for statement (ii) is closely related to that for statement (i). In the high-β equilibrium, the trader earns a lower trading profit due to a smaller informational advantage over the market maker. This result may appear inconsistent with statement (i) because, ceteris paribus, a lower price impact is typically associated with a higher trading profit. However, the trader's order size $|x|$ in the high-β equilibrium is so large that the *overall* price impact (i.e., λ multiplied by $|x|$) is larger than that in the low-β equilibrium. Overall, the trader moves the price excessively and earns a lower trading profit in this equilibrium. Due to her aggressive trading, more information about s—which is informative about δ—is incorporated into the price in the high-β equilibrium, supporting statement (iii).

Statement (iv) follows, again, from the trader's aggressive trading in the high-β equilibrium. Specifically, she places a large buy (sell) order when $s > 0$ ($s < 0$), even if $|s|$ is small. This causes a large upward (downward) pressure on p, leading to greater price variability. This result aligns with the insight of Guerrieri and Kondor (2012) that reputation concerns amplify price volatility, although their mechanism differs from ours.[13]

In the high-β equilibrium, the trader reveals more information about s, thereby improving the precision of the recruiter's estimation (statement (v)). This makes the reputation benefit more responsive to skill, as the recruiter can assess it more accurately.

[13] Guerrieri and Kondor (2012) argue that bond price volatility increases because fund managers demand a premium to offset the risk of reputation damage in the event of default.

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